FINM3013 Introduction to Financial Derivatives

Chapter 6 Exercises

Solutions

1.

There are 176 days between February 4 and July 30 and 181 days between February 4 and August 4. The cash price of the bond is, therefore:

$$110+\frac{176}{181}\times 6.5=116.32$$

The rate of interest with continuous compounding is $2\ln 1.06=0.1165$ or 11.65% per annum. A coupon of 6.5 will be received in 5 days (= 0.01370 years) time. The present value of the coupon is

$$6.5e^{-0.01370\times 0.1165}=6.49$$

The futures contract lasts for 62 days (= 0.1699 years). The cash futures price if the contract were written on the 13% bond would be

$$(116.32-6.49)e^{0.1699\times 0.1165}=112.03$$

At delivery there are 57 days of accrued interest. The quoted futures price if the contract were written on the 13% bond would therefore be

$$112.03-6.5\times\frac{57}{184}=110.01$$

Taking the conversion factor into account the quoted futures price should be:

$$\frac{110.01}{1.5}=73.34$$

2.

a) On the first day of the delivery month the bond has 15 years and 7 months to maturity. The value of the bond assuming it lasts 15.5 years and all rates are 6% per annum with semiannual compounding is

$$\sum_{i=1}^{31}\frac{5}{1.03^i}+\frac{100}{1.03^{31}}=140.00$$

The conversion factor is therefore 1.4000.

b) On the first day of the delivery month the bond has 21 years and 4 months to maturity. The value of the bond assuming it lasts 21.25 years and all rates are 6% per annum with

semiannual compounding is

$$\frac{1}{\sqrt{1.03}}\left[3.5+\sum_{i=1}^{42}\frac{3.5}{1.03^i}+\frac{100}{1.03^{42}}\right]=113.66$$

Subtracting the accrued interest of 1.75, this becomes 111.91. The conversion factor is therefore 1.1191.

c) For the first bond, the quoted futures price times the conversion factor is
118.71875×1.4000 = 166.2063
This is 2.7938 less than the quoted bond price. For the second bond, the quoted futures price times the conversion factor is
118.71875×1.1191 = 132.8576
This is 3.1418 less than the quoted bond price. The first bond is therefore the cheapest to deliver.

d) The price received for the bond is 166.2063 plus accrued interest. There are 175 days between January 1, 2014 and June 25, 2014. There are 181 days between January 1, 2014 and July 1, 2014. The accrued interest is therefore

$$5\times\frac{175}{181}=4.8343$$

The cash price received for the bond is therefore 171.0406.

3.

The number of short futures contracts required is

$$\frac{100,000,000\times 4.0}{122,000\times 9.0}=364.3$$

Rounding to the nearest whole number 364 contracts should be shorted.

a) This increases the number of contracts that should be shorted to

$$\frac{100,000,000\times 4.0}{122,000\times 7.0}=468.4$$

or 468 when we round to the nearest whole number.

b) In this case the gain on the short futures position is likely to be less than the loss on the bond portfolio. This is because the gain on the short futures position depends on the size of the movement in long-term rates and the loss on the bond portfolio depends on the size of the movement in medium-term rates. Duration-based hedging assumes that the

movements in the two rates are the same.